ZOO501-Developmental-Biology

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Solved Questions FINAL TERM (current PAPERS) February2020

1. Polyspermy 2 marks

The entrance of multiple sperm polyspermy leads to disastrous consequences in most organisms. In the sea urchin, fertilization by two sperm results in a triploid nucleus, in which each chromosome is represented three times rather than twice (Worse condition). Species have evolved ways to prevent the union of more than two haploid nuclei. The most common way is to prevent the entry of more than one sperm into the egg.

polyspermy describes an egg that has been fertilized by more than one sperm. Diploid organisms normally contain two copies of each chromosome, one from each parent. The cell resulting from polyspermy, on the other hand, contains three or more copies of each chromosome—one from the egg and one each from multiple sperm. Usually, the result is an inviable zygote

2. Zona pellucida 2 marks

Zona pellucida: The strong membrane that forms around an ovum as it develops in the ovary. The membrane remains in place during the egg's travel through the fallopian tube. To fertilize the egg, a sperm must penetrate the thinning zona pellucida. If fertilization takes place, the zona pellucida disappears, to permit implantation in the uterus.

The **zona pellucida** supports communication between oocytes and follicle cells during oogenesis; protects oocytes, eggs, and embryos during development, and regulates interactions between ovulated eggs and freeswimming sperm during and following fertilization.

3. Marble bone disease 2 marks

Osteopetrosis or Marble bone disease

Osteopetrosis is a bone disease that makes bones abnormally dense and prone to breakage (fracture). This disorder is characterized by increased bone density and abnormal bone growth.

Marble bone disease, also called **osteopetrosis** or **Albers-Schönberg disease**, rare disorder in which the bones become extremely dense, hard, and brittle. The disease progresses as long as bone growth continues; the marrow cavities become filled with compact bone. Because increased bone mass crowds the bone marrow, resulting in a reduced amount of marrow and therefore a reduced capacity to produce red blood cells, severe anemia results

4. Bone precursor cell 2 marks

During intramembranous ossification in the skull, neural crest derived mesenchymal cells proliferate and condense into compact nodules. Some of these cells develop into capillaries; others change their shape to become osteoblasts, committed **bone precursor cells.**

What happens when fusion occur between sperm and egg 2 marks

Recognition of sperm by the vitelline envelope or zona pellucida is followed by the lysis of that portion of the envelope or zona in the region of the sperm head by the acrosomal enzymes. This lysis is followed by the fusion of the sperm plasma membrane with the plasma membrane of the egg. The sperm and egg plasma membranes then join together, and material from the sperm membrane can later be found on the egg membrane. The sperm nucleus and tail pass through the resulting cytoplasmic bridge, which is widened by the actin polymerization.

The process of fertilization involves a sperm fusing with an ovum. After the sperm enters the cytoplasm of the oocyte (also called ovocyte), the tail and the outer coating of the sperm disintegrate and the cortical reaction takes place, preventing other sperm from fertilizing the same egg. The oocyte now undergoes its second meiotic division producing the haploid ovum and releasing a polar body. The sperm nucleus then fuses with the ovum, enabling fusion of their genetic material. The successful **fusion** of gametes forms a new organism

5. Fertilin function 2 marks

- In mammals, the fertilin proteins in the sperm plasma membrane are essential for sperm membraneegg membrane fusion
- Mouse fertilin is localized to the posterior plasma membrane of the sperm head. It adheres the sperm to the egg by binding to the 6β1 integrin protein on the egg plasma membrane
- fertilin appears to bind the sperm plasma membrane to the egg plasma membrane and then to fuse the two of them together. When the membranes are fused, the sperm nucleus, mitochondria, centriole, and flagellum can enter the egg
- Fertilin is a sperm surface protein with an essential role in fertilization. It is required for the migration of spermatozoa through the oviduct, for binding to the zona pellucida, and for efficient binding to the egg plasma membrane

6. Function of dynein 2 marks

The core of the axoneme consists of two central microtubules surrounded by a row of nine doublet microtubules. The force for sperm propulsion is provided by dynein, a protein that is attached to the microtubules. Dynein hydrolyzes molecules of ATP and can convert the released chemical energy into the mechanical energy that propels the sperm.

7. Blastomere progress 2 marks

Blastomeres generally progress through a cell cycle consisting of just two steps: M (mitosis) and S (DNA synthesis).

8. Embryonic shield 2 marks

The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the embryonic shield.

Also called **embryonic shield**. in the early **embryo** of mammals, the flattened inner cell mass that arises at the end of the blastocyst stage and from which the **embryo** begins to differentiate.

9. Example of enhancer in human 2 marks

The human β-globin gene has an enhancer in its 3´ UTR, roughly 700 base pairs

10. What is homology of Hox genes in droshophila and of mammals 2 marks

The homology of gene structure and the similarity of expression patterns between Drosophila and mammalian Hox genes suggests that this patterning mechanism is extremely ancient.

11. Process of neural crest

The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types. **The Neural Crest**

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.The neural crest cells originate at the dorsal most region of the neural tube.The neural crest can be divided into four main functional (but overlapping) domains.

The cranial (cephalic) neural crest, whose cells migrate dorsolaterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia,and connective tissues of the face. **Major pathways in differentiation of neural crest**

The trunk neural crest, whose cells take one of two major pathways.

- Neural crest cells that become the pigment-synthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly.
- The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome.
- Sclerotomes are blocks of mesodermal cells, derived from somites, that will differentiate into the vertebral cartilage of the spine.
- The vagal and sacral neural crest, whose cells generate the parasympathetic (enteric) ganglia of the gut.

- The vagal (neck) neural crest lies opposite chick somites 1-7, while the sacral neural crest lies posterior to somite 28.
- The cardiac neural crest is located between the cranial and trunk neural crests.
- The cardiac neural crest cells can develop into melanocytes, neurons, cartilage, and connective tissue (of the third, fourth, and sixth pharyngeal arches).

12. Causes in the activity cyclin of MPF

Mitosis-promoting factor contains two subunits.

The large subunit is called cyclin B.
Small subunit of MPF, the cyclin-dependent kinase

Cyclin B is often encoded by mRNAs stored in the oocyte cytoplasm, and if the translation of this message is specifically inhibited, the cell will not enter mitosis. The presence of cyclin B depends upon its synthesis and its degradation. Cyclin B regulates the small subunit of MPF, the **cyclin-dependent kinase**.

13. Difference between Enhancer and Promoter?

An **enhancer** is a sequence of DNA that functions to enhance transcription. A **promoter** is a sequence of DNA that initiates the process of transcription. A **promoter** has to be close to the gene that is being transcribed while an **enhancer** does not need to be close to the gene of interest.

14. Mode of commitment

Three basic modes of commitment are,

❖ Autonomous specification
❖ Conditional specification ❖ Syncytial specification

15. What is sea urchin larva name 2 marks

Pluteus larve

16. Cleavage in sea urchin

As the 16-cell embryo cleaves, the eight mesomeres divide to produce two "animal" tiers, an1 and an2, one staggered above the other. The macromeres divide meridionally, forming a tier of eight cells below an2.

17. How neural groove form

Shortly after the neural plate has formed, its edges thicken and move upward to form the neural folds, while a U-shaped neural groove appears in the center of the plate, dividing the future right and left sides of the embryo.

18. Retinoic acid foundation 3 marks

The specification of the ventral neural tube appears to be mediated by external tissues.One agent of ventral specification is the Sonic hedgehog protein, probably originating from the notochord. Another agent specifying the ventral neural cell types is retinoic acid, which probably comes from the adjacent somites.

19. What is hierarchical development 3 marks

The developmental hierarchy is the hierarchical series of decisions involving specific preexisting cell types.
These developmental decisions are usually irreversible, that progressively and irreversibly restrict cell fate.
Animal development is the division of embryo into the three germ layers. 1. Ectoderm
2. Mesoderm
3. Endoderm

20. periosteum 3 marks

the entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the Periosteum (a membrane that surrounds the bone).

The **periosteum** is a membrane that covers the outer surface of all bones, except at the joints of long bones. Endosteum lines the inner surface of the medullary cavity of all long bones.

21. Difference between germ cells and somatic cells 5 marks

In many species a specialized portion of egg cytoplasm gives rise to cells that are the precursors of the gametes (the sperm and egg). The gametes and their precursor cells are collectively called germ cells, and they are set aside for reproductive function.

All the other cells of the body are called Somatic cells. This separation of somatic cells (which give rise to the individual body) and germ cells (which contribute to the formation of a new generation) is often one of the first differentiations to occur during animal development.

22. Difference between fraternal or identical twins 5 marks

Fraternal twins are the result of two separate fertilization events, whereas identical twins are formed from a single embryo whose cells somehow dissociated from one another.

Identical twins occur in roughly 0.25% of human births. About 33% of identical twins have two complete and separate chorions, indicating that separation occurred before the formation of the trophoblast tissue at day 5.

23. Primary neurulation and secondary neurulation 5 marks

In primary neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

During primary neurulation, the original ectoderm is divided into three sets of cells.
▪ The internally positioned neural tube, which will form the brain and spinal cord. ▪ The externally positioned epidermis of the skin.
The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

In secondary neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary.

Secondary neurulation involves the making of a medullary cord and its subsequent hollowing into a neural tube. In frogs and chicks, secondary neurulation is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae. In both cases, it can be seen as a continuation of gastrulation. The growing region at the tip of the lip is called the chordoneural hinge, and it contains precursors for both the posterior most portion of the neural plate and the posterior portion of the notochord. The tip of the tail is the direct descendant of the dorsal blastopore lip, and the cells lining the blastopore form the neurenteric canal.

24. Three axes of body foundation 5 marks

Three axes are the foundations of the body:

- the anterior-posterior axis (head to tail or mouth to anus),
- the dorsal-ventral axis (back to belly),
- the right-left axis (between the two lateral sides of the body).

25. Step of sperm and egg interaction 5 marks

The interaction of sperm and egg generally proceeds according to five basic step:

➢ The chemo-attraction of the sperm to the egg by soluble molecules secreted by the egg.
➢ The exocytosis of the acrosomal vesicle to release its enzymes.
➢ The binding of the sperm to the extracellular envelope (vitelline layer or zona pellucida) of the egg.
➢ The passing of the sperm through this extracellular envelope.
➢ Fusion of egg and sperm cell plasma membranes.

26. Role of calcium in cortical granule 5 marks

Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents. Once the

fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg. The release of calcium from intracellular storage can be monitored visually using calcium activated luminescent dyes such as aequorin (isolated from luminescent jellyfish) or fluorescent dyes such as fura-2. These dyes emit light when they bind free calcium ions.

Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

27. Structure of sperm 5 marks

Each sperm consists of,

1) A haploid nucleus,
2) A propulsion system to move the nucleus, and
3) A sac of enzymes that enable the nucleus to enter the egg.

During the course of sperm maturation, the haploid nucleus becomes very streamlined, and its DNA becomes tightly compressed

In front of this compressed haploid nucleus lies the acrosomal vesicle, or acrosome, which is derived from the Golgi apparatus and contains enzymes that digest proteins and complex sugars; thus, it can be considered a modified secretory vesicle.

These stored enzymes are used to lyse the outer coverings of the egg. In many species, such as sea urchins, a region of globular actin molecules lies between the nucleus and the acrosomal vesicle.

These proteins are used to extend a fingerlike acrosomal process from the sperm during the early stages of fertilization.

28. Different steps of cleavage in development of snail 10 marks

Spiral holoblastic cleavage is characteristic of several animal groups, including annelid worms, some flatworms, and most molluscs. It differs from radial cleavage in numerous ways..

First, the cleavage planes are not parallel or perpendicular to the animal-vegetal axis of the egg; rather, cleavage is at oblique angles, forming a "spiral" arrangement of daughter blastomeres
Second, the cells touch one another at more places than do those of radially cleaving embryos. In fact, they assume the most thermodynamically stable packing orientation, much like that of adjacent soap bubbles.
Third, spirally cleaving embryos usually undergo fewer divisions before they begin gastrulation, making it possible to follow the fate of each cell of the blastula.
Blastulae produced by radial cleavage have no blastocoel and are called stereoblastulae

The first two cleavages are nearly meridional, producing four large macromeres (labeled A, B, C, and D). In each successive cleavage, each macromere buds off a small micromere at its animal pole

29. Neurlation defects 10 marks

Different neural tube defects are caused when various parts of the neural tube fail to close. Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed. Failure to close the anterior neural tube regions results in a lethal condition, anencephaly. Here, the forebrain remains in contact with the amniotic fluid and subsequently degenerates. Fetal forebrain development ceases, and the vault of the skull fails to form. The failure of the entire neural tube to close over the entire body axis is called Craniorachischisis. Collectively, neural tube defects are not rare in humans, as they are seen in about 1 in every 500 live births

30. Cleavage in mammals 10 marks

The mammalian oocyte is released from the ovary and swept by the fimbriae into the oviduct. Fertilization occurs in the ampulla of the oviduct, a region close to the ovary. Meiosis is completed at this time, and first cleavage begins about a day later. Cleavages in mammalian eggs are among the slowest in the animal kingdom about 12-24 hours apart. Meanwhile, the cilia in the oviduct push the embryo toward the uterus; the first cleavages occur along this journey.

The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction

The cells of the compacted 8-cell embryo divide to produce a 16-cell morula. The morula consists of a small group of internal cells surrounded by a larger group of external cells.

Most of the descendants of the external cells become the trophoblast (trophectoderm) cells. This group of cells produces no embryonic structures. Rather, it forms the tissue of the chorion, the embryonic portion of the placenta.

These cells generate the inner cell mass (ICM), which will give rise to the embryo and its associated yolk sac, allantois, and amnion. By the 64-cell stage, the inner cell mass (approximately 13 cells) and the trophoblast cells have become separate cell layers, neither contributing cells to the other group. During a process called cavitation, the trophoblast cells secrete fluid into the morula to create a blastocoel. The inner cell mass is positioned on one side of the ring of trophoblast cells. The resulting structure, called the blastocyst, is another hallmark of mammalian cleavage.

Once out, the blastocyst can make direct contact with the uterus. The uterine epithelium (endometrium) "catches" the blastocyst on an extracellular matrix containing collagen, laminin, fibronectin, hyaluronic acid, and heparin sulfate receptors

31. What is cleavage 2 marks

Cleavage is a series of extremely rapid mitotic divisions wherein the enormous volume of zygote cytoplasm is divided into numerous smaller cells. These cells are called blastomeres, and by the end of cleavage, they generally form a sphere known as a blastula.

32. Define isolecithal 2 marks

At one extreme are the eggs of sea urchins, mammals, and snails. These eggs have sparse, equally spaced yolk and are thus isolecithal (Greek, "equal yolk")

33. Patterns of embryonic cleavage 10 marks

Indeed, different organisms undergo cleavage in distinctly different ways. The pattern of embryonic cleavage particular to a species is determined by two major parameters:

1. The amount and distribution of yolk protein within the cytoplasm.
2. The factors in the egg cytoplasm that influence the angle of the mitotic spindle and the timing of its formation.
3. The yolk-rich pole is referred to as the vegetal pole; the yolk concentration in the animal pole is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.

34. What is fast block to polyspermy

The Fast block to polyspermy is achieved by changing the electric potential of the egg plasma membrane. This membrane provides a selective barrier between the egg cytoplasm and the outside environment, and the ionic concentration of the egg differs greatly from that of its surroundings. This concentration difference is especially significant for sodium and potassium ions. Seawater has a particularly high sodium ion concentration, whereas the egg cytoplasm contains relatively little sodium. The reverse is the case with potassium ions.

34. Difference between dextral coiling and sinistral coiling.

Some snails have their coils opening to the right of their shells (dextral coiling), whereas other snails have their coils opening to the left (sinistral coiling).

35. Four level of gene expression

The regulation of gene expression can be accomplished at several levels:

Differential gene transcription:

It regulates that which of the nuclear genes are transcribed into RNA.

Selective nuclear RNA processing:

It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.

Selective messenger RNA translation:

It regulates that which of the mRNAs in the cytoplasm become translated into proteins.

Differential protein modification:

It regulates that which proteins are allowed to remain or function in the cell.

36. Explain the spina bifida

Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed.

37. What is cone growth

Dendrites are often numerous and do not extend far from the neuronal cell body, or soma, axons may extend for several feet.The Nerve outgrowth is led by the tip of the axon, called the growth cone. The growth cone moves by the elongation and contraction of pointed filopodia called Microspikes. These microspikes contain microfilaments, which are oriented parallel to the long axis of the axon.

38. What is hypoblast

Area pellucida cells migrate individually into the subgerminal cavity to form the polyinvagination islands (primary hypoblast). Shortly thereafter, a sheet of cells from the posterior margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the polyinvagination islands, thereby forming the secondary hypoblast.

39. Meta cephalon 2 marks

The Metencephalon gives rise to the cerebellum, the part of the brain responsible for coordinating movements, posture, and balance.

40. Transcription factor

Transcription factors domains

Transcription factors have three major domains.

1) DNA-binding domain
2) Protein-protein interaction domain
3) Trans-activating domain

DNA-binding domain:

It recognizes a particular DNA sequence.

Protein-protein interaction domain:

It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.

Trans-activating domain:

It activates or suppresses the transcription of the gene whose promoter or enhancer it has bound. Usually, this trans-activating domain enables the transcription factor to interact with proteins involved in binding RNA polymerase.

Examples of Transcription factors

MITF:
(Microphthalmia-Associated Transcription Factor. The microphthalmia (MITF) protein is necessary for the production of pigment cells and their pigments. **Pax6:**
Paired box gene 6. The Pax6 transcription factor, which is needed for mammalian eye, nervous system, and pancreas development, contains two potential DNA-binding domains. **Pdx1:**
Pancreatic And Duodenal Homeobox 1 gene. It is specific for the pancreatic region of the endoderm.

41. Cortice and nuclei 2 marks

In the cerebellum, some neuronal precursors enter the marginal zone to form clusters of neurons called nuclei Each nucleus works as a functional unit, serving as a relay station between the outer layers (cortices) of the cerebellum and other parts of the brain

42. Types of cleavage in snail 2 marks

At one extreme are the eggs of sea urchins, mammals, and snails. These eggs have sparse, equally spaced yolk and are thus isolecithal (Greek, "equal yolk"). In these species, cleavage is holoblastic (Greek holos, "complete"). meaning that the cleavage furrow extends through the entire egg.

Zygotes containing large accumulations of yolk undergo meroblastic cleavage, wherein only a portion of the cytoplasm is cleaved. The cleavage furrow does not penetrate into the yolky portion of the cytoplasm.

The eggs of insects have their yolk in the center (i.e., they are centrolecithal), and the divisions of the cytoplasm occur only in the rim of cytoplasm around the periphery of the cell (i.e., superficial cleavage).

The eggs of birds and fishes have only one small area of the egg that is free of yolk (telolecithal eggs), and therefore, the cell divisions occur only in this small disc of cytoplasm, giving rise to the discoidal pattern of cleavage.

43. Events in primary neurulation 5 marks

During primary neurulation, the original ectoderm is divided into three sets of cells.

- The internally positioned neural tube, which will form the brain and spinal cord.
- The externally positioned epidermis of the skin.

The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types

44. What type of molecules involved in histone protein.2 marks

Nucleosome is composed of an octamer of histone proteins (two molecules each of histones H2A-H2B and histones H3-H4) wrapped with two loops containing approximately 140 base pairs of DNA.

45. Difference b/w Karyokinesis and Cytokinesis 5 marks

karyokinesis the mitotic division of the nucleus.

The mechanical agent of this division is the mitotic spindle, with its microtubules composed of tubulin (the same type of protein that makes up the sperm flagellum).

The second process is **cytokinesis** the division of the cell.

The mechanical agent of cytokinesis is a contractile ring of microfilaments made of actin (the same type of protein that extends the egg microvilli and the sperm acrosomal process).

The relationship and coordination between these two systems during cleavage is depicted in Figure, in which a sea urchin egg is shown undergoing first cleavage.

46. meroblastic

The cell divisions do not completely divide the egg, so this type of cleavage is called **meroblastic**.

47. Dose ascidians have notochord 2 marks

Ascidians, members of the tunicate subphylum, are fascinating animals for several reasons, but the foremost is that they are invertebrate chordates. They have a notochord as larvae (and therefore are chordates), but they lack vertebrae.

48. What BHLH protein specific for. 2 marks

the myogenic bHLH proteins are specific for muscle cells. Any cell making a myogenic bHLH transcription factor such as MyoD or Myf5 is committed to becoming a muscle cell.

49. What is promoter 3 marks

Promoter:
A region, which is responsible for the binding of RNA polymerase and for the subsequent initiation of transcription.

50. Define air space 3 marks

An air space forms when the contents of the egg cool and contract after the egg is laid. The air cell usually rests between the outer and inner membranes at the egg's larger end.

51. Describe osteoclast in detail 5 marks

Osteoclasts are probably derived from the same precursors as macrophage blood cells, and they dissolve both the inorganic and the protein portions of the bone matrix.
Each osteoclast extends numerous cellular processes into the matrix and pumps out hydrogen ions onto the surrounding material, thereby acidifying and solubilizing it.
The blood vessels also import the blood-forming cells that will reside in the marrow for the duration of the organism's life. The number and activity of osteoclasts must be tightly regulated.
If there are too many active osteoclasts, too much bone will be dissolved, and osteoporosis will result. Conversely, if not enough osteoblasts are produced, the bones are not hollowed out for the marrow, and osteopetrosis results.

52. Write classes of cadherins 5 marks

In vertebrate embryos, several major cadherin classes have been identified:

1. E-cadherin
2. P-cadherin
3. N-cadherin
4. Protocadherins

E-cadherin: a. Epithelial cadherin is expressed on all early mammalian embryonic cells, even at the 1cell stage. Later, this molecule is restricted to epithelial tissues of embryos and adults.

P-cadherin: a. Placental cadherin appears to be expressed primarily on the trophoblast cells (those placental cells of the mammalian embryo that contact the uterine wall) and on the uterine wall epithelium.

N-cadherin: a. Neural cadherin is first seen on mesodermal cells in the gastrulating embryo as they lose their E-cadherin expression. It is also highly expressed on the cells of the developing central nervous system.

Protocadherins: a. Calcium-dependent adhesion proteins that differ from the classic Cadherins in that they lack connections to the cytoskeleton through catenins. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during Xenopus gastrulation.

53. Definer silencer 2 marks

Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "negative enhancers " are also called silencers.

54. Difference between sex and reproduction 5 marks

Sex and reproduction are two distinct and separable processes. Reproduction involves the creation of new individuals. Sex involves the combining of genes from two different individuals into new arrangements.
Sex and reproduction are two different processes that can or not to occur simultaneously. Sex must be considered as an exchange of genetic material doing new combinations, that can originate or not new individuals. While that the reproduction has that obligatorily to originate new beings.

55. Differentiation process of medulla and spinal chord 10 marks

As the cells adjacent to the lumen continue to divide, the migrating cells form a second layer around the original neural tube. This layer becomes progressively thicker as more cells are added to it from the germinal neuroepithelium. This new layer is called the mantle (or intermediate) zone, and the germinal epithelium is now called the ventricular zone (and, later, the ependyma).
The mantle zone cells differentiate into both neurons and glia. The neurons make connections among themselves and send forth axons away from the lumen, thereby creating a cell-poor marginal zone. Eventually, glial cells cover many of the axons in the marginal zone in myelin sheaths, giving them a whitish appearance. Hence, the mantle zone, containing the neuronal cell bodies, is often referred to as the Gray matter; the axonal, marginal layer is often called the White matter. In the spinal cord and medulla, this basic three-zone pattern of ependymal, mantle, and marginal layers is retained throughout development. The gray matter (mantle) gradually becomes a butterfly-shaped structure surrounded by white matter; and both become encased in connective tissue.
As the neural tube matures, a longitudinal groove the sulcus limitans divides it into dorsal and ventral halves. The dorsal portion receives input from sensory neurons, whereas the ventral portion is involved in effecting various motor functions.

Z00501 FINAL TERM MCQS

1. Mode of commitment
2. strageries of commitment
3. Rana pipiens usually lays around 2500 eggs while the bullfrog, Rana catesbiana, can lay as many as 20 000
4. Multinucleated type of cell form.
5. slow block, initiated by the intracellular release calcium ions of
6. There are two major modes of bone formation, or osteogenesis
7. Mutations of the SOX9 gene cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body
8. Human brain contains 10^{12} glial cells
9. Telolecithal is cleavage is found in eggs of Birds and Fishes
10. Pax6 plays a major role in eye formation
11. Pax1 and Scleraxis are the transcription factors are thought to activate cartilage-specific genes
12. The eggs of insects center have their yolk in the(i.e., they are centrolecithal),
13. Egg more Ca activates due to which ion more
14. Which n Ectoderm dermis give rise to brain
15. Which Ectoderm dermis give rise to eye
16. Mod of syncvtial specification commitment in insects
17. Regulative development arises from conditional specification
18. Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately 140 base pairs of DNA
19. the membrane potential of the more negative sperm becomesas potassium ions leave the sperm
20. Paracrine factors such as TGF- and FGF7 are skin development important in normal
21. frog The male leopard release sperm in the summer
22. In zona pellucid mouse ZP3 i s attached to zo
23. It is estimate egg is 25000 Sea Urchin to 50000 mRNA.
24. Fertilin protein is important for sperm-egg fusion in Mammals
25. Fast block to Sodium polyspermy is initiated by
26. Transition from factor MPF fertilization to cleavage is caused by factor isolethial
27. Eggs having sparse and equally yolk space are called
28. primitive gut Archenteron is used for
29. Spiral cleavage found in

30. Sonic Snails hedgehog is a type of

31. Which of protein type of neurulation in fish

32. Nural n in fish secondary crest divides functionally in domains

onally in domains 4

muscle cell 33. Myogenic bHLH proteins are specific for development of

Zygote which d

Tatratology

34. which divide mitotically to produce all the cells of the body.

35. the study of birth defect.

36. Identical twins occur in 0.25% roughlyof human births.

37. Requlatively development is seen in most vertebrate embroys , and it is obviously critical in development of identical twins.

38. Insect such as drosophila use all three mods of specification to commit their cells to particular fates.

39. speed of metamorphosis is directly linked with environment

In 40. acetabularia unicellular organisms, development us controlled at both the transciptional and translational

41 The individuals of each species, however is divided into two mating types Plus and minus calamadomonas.

42. 140 base pairs nucleosome. of DNA

43. envelop contains at least eight different glycoproteins.

44. first evidence suggestion the importance of sperm in reproduction came from Lazzaro Spallanzani a series experiment performed by

45. The volume of the sea urchin is 200 aboutPico liters

46. Hindbrain Rhombencephalon

47. most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction

48. embryo contains three germ layers. The ectoderm, the endoderm and the mesoderm.

49. neural form keel fish

50. holoblastic clevage is characteristic of several animals groups, including annelid worms, some flateworms, and most mollusks(snails).

Diploblastic animals 51are those who have ectoderm and endoderm but no true mesoderm (hydra)

52. fate is determined by signalling. The micromeres constitute a major signalling center. B-catenin is important for the inducing capacity of micromeres.

53. Even distribution of yolk in isolecithal ovum

54. of bindin protein 30 500-Da

55. which has two parts in cell cycle Blastomere options

germ ring

56. After the blastoderm cells have covered about half the zebra fish _____ yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the

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, and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not

64. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the mesodermal somites in the fish.

65. The rudiment of the central nervous system, is called neurulation.

66. Blastula produced by radial cleavage has no blastocoel and are called stereoblastulae.

____plate_____indu

67. Fraternal twins are the result of two separate fertilization events.

68. The prechordal induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord.

69. The resulting larvae of snail is called glochidia which resemble tiny bear traps.

Z00501 CURRENT FINAL TERM MCQS

1. Due to Loss in hydrogen ions, pH rises.
2. As a result of gastrulation, ectoderm, mesoderm and Endoderm are formed.
3. Male leopard Frogs make their sperm in summer.
4. Differential gene transcription is step for gene expression regu lation.
5. Coding regions of DNA exon.
6. How many domains are present for transcription 3
7. pdxl is present in Pancreatic And Duodenall gene.
8. Bacteria transmit genes from one cell to other by sex pili.
9. Cells rearrange their DNA to create new immunoglobin and antigen receptor gene.
10. Sea urchins contain 25 000 to 50 000 mRNA.
11. In many species, sperm are attracted towards eggs by chemotaxis.
12. Acrosomal protein mediating recognition by proteinbindin
13. Sea water has High sodium ion conc.
14. Eggs have sparse equally spaced yolk are icole
15. Cytoplasm of blastodisc become embr is mero division called discoidal.
16. The micromere constitute major na ce
17. Osteoblasts secrete collage te ycan matrix that is able to bind calcium salts
18. Tgf beta family secr d the surface ectoderm and roof of neural tu
19. Dendrites n and do not extend far from the dy soma axons may extend for several feet.
20. One agent o ventral specification is sonic hedgegog protein.
21. In frogs and chicks, secondary nueralation is seen in neural tube.
22. The cells at dorsal most portion of neural tube become neural crest cells.
23. About 33% of identical twins have two complete and separate chorions
24. Mammals undergo holoblastic rotational va
25. In birds, gray.jgvis critical in d termi osterior axis
26. Decidua becomes richod els t at will provide oxygen and nut sto b
27. Mammalian cyt r from ovary and swept by fimbriae i .t.b.e—C- y-LUC.t-:
28. Fertl •ck g occurs in oviduct.
29. lop tal decisions are usually irreversible.
30. A ar organisms arise by a slow process of ro ssive change called development.
31. jor class of cadherin includes N- cadherin.

Only small % of genome is expressed in partial cell.

33. Cortical are granules are membrane bound structure containing proteolytic enzymes.
34. Small subunits of MPF cyclin-dependent kinase.
35. It secretes a cellulose tunic (which gave the name "tunicates" to these creatures).
36. Inner surface cells thin 30x5um process filopodia.
37. ß-catenin is Important for Inducing capacity of micromeres.
38. In fishes , First divisions occur synchronously form a mouln o cells that sits an animal pole of large yolk cell.
39. Fertilization occurs in ampulla of oviduct a close, a region

close to ovary.

40. In amphibians most of the xenopus most of tadpole is formed by primary neurulation and tail of neural tube formed by 2ndry neuralation.

41. Neural fold migrates to midline of embryo and fusing to

42. Tissues form nervous system in the fly or in the frog.

43. The marginal zones to form clusters of neurons called nuclei.

44. Paracrine factors instruct Mvotome cells to become muscles by inducing them to synthesize MyoD protein.

45. Sclerix is expressed in mesenchyme from the scleretome.

46. Acrosomal reaction in sea urchins is often highly specific to each species.

47. Cells are arrange their.. to creat new immunoglobin

48. In frogs and chicks, secondary neurulation is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae.

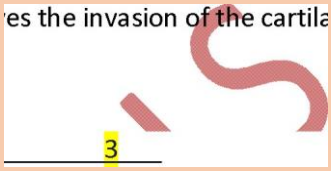
65. This fetal organ the Chorion is derived primarily from embryonic trophoblast cells.

66. Secondary neurulation involves the making of a medullarv cord and its subsequent hollowing into neural tube

67. The mantle zone cells differentiate into both neurons and

68. The cerebral cortex in humans has six layers.

69. Paracrine factors such as TGF and FGF7 are important in normal skin development.

70. The fifth phase 

involves the invasion cartilage model by blood vessels.

3

71. Mode of commitment

72. strageries of commitment 2

73. Rana pipiens usually lays around 2500 eggs while the bullfrog, Rana catesbiana, can lay as many as 20 000

74. Multinucleated type of cell form. syncvtiotrophoblast

49 Regulation of gene expression by **several level.**

50. Which type of neuralation in fish **exclusively secondary**.

51 sonic hedgehocy is a type of **protein.**

52 **Blastomeres** have two parts of cell cycle.

53 Number of cortical granules Sea urchin **15000.**

54 How many strategies involve in cell commitment **2.**

55 Resting membrane potential of the egg is **70mv.**

56 One chemotactic molecule, a 14-amino acid peptide called **resact**.

57 Force for sperm propulsion is provided by **dynin.**

58 Regulative development arises from **Conditional Specification.**

59 Human brain contains **10^{12} glial cells.**

60 Fish have discoidal **telolicithal** pattern.

61 Process of osteogensis **2**

62. The mitotic spindle and contractile ring are **perpendicular** to each other.

63 Gastrulation in Snails is accomplished primarily by **epiboly.**

64 The first cell movement of fish gastrulation is the **epiboly** of the blastoderm.

75. slow block, initiated by the intracellular release of calcium ions

79. Telolecithal is cleavage is found in eggs of Fishes

Pax6 plays a major role in eye formation and Scleraxis are the transcription factors are thought to activate cartilage-specific genes

Pax1 an

The eggs of insects have their yolk in the (i.e., they are centrolecithal),

Egg activates due to which ion more

Which dermis give rise to brain

Which dermis give rise to eye

Mod of commitment in insects

Regulative development arises from conditional

75. slow block, initiated by the intracellular release of calcium ions

76. There are two major modes of bone formation or **osteogenesis**

77. Mutations of the *SOX9* gene cause camptomelic dysplasia,

78. Human brain contains **10^{12}** glial cells

79. Telolecithal is cleavage is found in eggs of Birds and Fishes

80. Pax6 plays a major role in eye formation

81. Pax1 and Scleraxis are the transcription factors are thought to activate cartilage-specific genes

82. The eggs of insects have their yolk in the center (i.e., they are centrolecithal),

83. Egg activates due to which ion more Ca

84. Which dermis give rise to brain Ectoderm

85. Which dermis give rise to eye Ectoderm

86. Mod of commitment in insects syncytial specification

87. Regulative development arises from conditional specification

88. Nucleosome is composed of an octamer of histone protein wrapped with two loops containing approximately 140 base pairs of DNA

89. the membrane potential of the sperm becomes more negative as potassium ions leave the sperm

Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately of DNA

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90. Paracrine factors such as TGF- and FGF7 are important in normal skin development

91. In mouse ZP3 is attached to zona pellucid

92. It is estimate egg is 25000 to 50000 mRNA. Sea Urchin

93. Fertilin protein is important for sperm-egg fusion in Mammals

94. Fast block to polyspermy is initiated by Sodium

95. Transition from fertilization to cleavage is caused by factor MPF

96. Eggs having sparse and equally yolk space are called isolethial

97. Archenteron is used for

Join VU BIO

98. Spiral cleavage found in primitive gut

99. Sonic hedgehog is a Snails type of

e of protein

ation in fish secondary

nctionally in domains 4

1. Which type of neurulation in fish

2. Nural crest divides functionally in domains

3. Myogenic bHLH proteins are specific for development of muscle cell

Zygote 4.which divide mitotically to produce all the cells of the body.

Tatratology 5. the study of birth defect.

6. Identical twins occur in roughly 0.25 of human births.

7. Requlatively development is seen in most vertebrate embroys , and it is obviously critical in development of identical twins.

drosophila 8. Insect such as use all three mods of specification to commit their cells to particular fates.

pressures 9. The speed of metamorphosis is directly linked with environment

10. acetabularia unicellular organisms, development us controlled at both the transciptional and translational level.

11. The individuals of each species, however is divided into two mating types Plus and minus calamadomonas.

2. base pairs of ONA nucleosome.

3. envelop contains ontains at least eight different glycoproteins.

4. first evidence suggestion the importance of sperm in reproduction came from a series experiment performed by Lazzaro Spallanzani

5. The volume of the sea urchin is 200 about Pic

16. Hindbrain Rhombencephalon

17. most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction

18. embryo contains three germ lavers. The ectoderm, the endoderm and the mesoderm.

19. neural form keel fish

20. holoblastic clevage is characteristic of several animals groups, including annelid worms, some flatworms, and most mollusks (snails).

Diploblastic animals 21are those who have ectoderm and (hydra) endoderm but no true mesoderm

isolecithal 22. Even distribution of yolk in ovum

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glochidia 38. The resulting larvae of snail is called which resemble tiny bear traps.

39. D

40. Fertilization in the absence of sex occur through the Binary Fission

41. The macromeres divide meridionally, forming a tier of eight (8) cells

42. cells, derived from the micromeres, are called the primary mesenchvme.

43. Molluscus follow which type of cleavage holoblastic cleavage

44. A mammalian blastula is called a blastocyst.

45. Which of the following undergo discoidal cleavage fishes, reptiles and birds

46. The egg jelly factors that initiate the Acrosomal reaction in sea urchins are often highly specific to each species

47. The Metencephalon gives rise to the cerebellum

48. In primary neurulation, the original ectoderm is divided into three sets of cells

49. The growing region at the tip of the lip is called the chordoneural hinge

50. Neural Tube will form Brain and s inal cord

51. Each ICM (Inner Cell Mass) cell has the same p

52. Variable cleavages produce no invariant fate assignments to cells

53. In early embryos of these insects Il divi n is not complete

54. The neural crest cells origina e dorsal most region of the neural tub

55. The human as an enhancer in its 3' UT R, roughly 700 a a i

56. t SO gene hich encodes a DNAbinding protein

57. the e e r on of calcified spicules becomes surrou d by compact mesenchymal cells that form the Periosteum

58. The second step is the alignment of the myoblasts formation of the midbrain, hindbrain, and spinal cord. together into chains

59. Cleavages sphere is known as Blastula

60. Osteoclasts are probably derived from the same precursors as macrophage blood cells

61. charby desired to know the cause of birth defects

62. The direct conversion of mesenchymal tissue into bone i called intramembranous ossification

63. Those cells that remain integral co ents of the neural tube lining become e end mal

64. The original type of tro hoblast ells consti a layer called the cytotrophobl

65. The major structu hara e avian, reptilian, & mammalian gastrul rimitive streak

66. The Bias o c e is overned by the synthesis and ra ion

ha nu s and Toad and frog

68. Th av tochord as larvae (chordates), but they ve rae.

69. crosomal vesicles is derived from Enzymes Golgi apparatus and ntains

The large subunit of MPF is cyclin B

71. ZP3 is the specific glycoprotein in the mouse

72. MIT F necessa ry for the production of Pigment cells

73. Mutation of sox9 gene cause camptomelic dvsplasia

74. Between the blastoderm and the yolk is a space called the

75. The second step of gastrulation involves the involution of the mesoderm

76. The first cleavage furrow appears centrally in the

blastodisc

77. Fertilization occurs in the ampulla of the oviduct
78. Mother provides fetus with nutrients and oxygen
79. The first segregation of cells within the inner cell mass results in the formation of the hypoblast (sometimes called the primitive endoderm)
80. The fetal portion of placenta formed by Chorion
81. Paracrine factor instruct myotome cells to become

Muscles

82. When myoblasts stop dividing, secrete fibronectin
83. Retinoic Acid comes from 7.

Somites

84. Hox genes pattern the anterior-posterior axis and help to specify positions along that axis
85. The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and hypothalamic brain regions

8.

9. The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by in the late 1700s. Lazzaro Spallanzani

The early embryonic cells need a supply of energy and amino acids which is provided by voll« proteins in the

The depression is called the primiti groove

86. Endochondral ossification involves the formation of cartilage
87. J.Hammerling exchanged nuclei b/w two distinct species
88. When plus individuals initiate fusion by extending a fertilization tube. Which contacts & fuses with a specific site on the minus individual.
89. Promoter is enhancer which is thought to regulate transcription.
90. Identical twins share a common Chorion
91. In Birds, pH differences appear crucial for distinguishing dorsal from ventral
92. Which regulate function of Promoter Enhancer
93. The space between two neurons, where tr itt synapse released
94. integral components of the neural tube lim co ependvmal
95. Mantle cells differentiate int Gray matter
96. The peripheral ring of blast e ells th have not shed their deep cells the opaca.
97. During fertilization, the actin cules fuse to form the long microfilame

98. A conditio in ce Ck the instructions for develop

1. ta pho is i naive itiated by hormones from the 's thyroid gland
   100 billion neurons
3. Release of Ca from intracellular storage controlled by dye Aequorin
4. Nervous system develop by Neural Tube
5. Allows the mRNA to exit the nucleus poly a tail

Z00501 CURRENT MCQS MIDTERM (2016) UPDATED

1. Red blood cells make Globins
2. how many stra r velopmental commitment
   2
   1012
   sea urchin
3. Ribosomro
   urchin o egg jelly of
5. is at studies embryonic & other pm al
6. %age of genome expressed in particular cell

Di tive & respiratory system formed by Endoderm

8. ogesterone work for signals the egg to resume its meiotic division
9. Insect characteristics are specification

syncvtial

10. In Chlamydomonas haploid chromosome has.....copy
one

11. Modes of commitments are..... 3

12. After fertilization developmental process proceed by
Cleavage

13. Ca ion responsible for cortical granules stores in
Endoplasmic Reticulum of Eggs

14. Seawater has a particularly high ion concentration
Na (Sodium)

15. There are approaches of anatomy 4(four)

16. The major cell adhesion molecules appear to be the
Cadherins.

17. It is estimate egg is 25000 to 50000 mRNA.
Sea Urchin

18. Fusion of and sperm cell plasma membrane.
Egg

19. Resact diffuse readily in sea water.

20. The requirement of capacitance varies from species to

6. Contact and between sperm and egg Recognition Species.